

PARTIAL ASSOCIATION EXPLORER

GLOBAL AND LOCAL (PARTIAL) ASSOCIATIONS IN MIXED-TYPE DATA

1 - Introduction and Workflow

- Partial Association Explorer** is an open-source **R Shiny application** for profiling unconditional and conditional associations in multivariate datasets containing **numerical and categorical variables**.
- It builds on and extends *AssociationExplorer* [SoftwareX] by introducing conditional association analysis, significance tests, new pair-plot visualizations, and likelihood-based measures for categorical pairs.
- For every selected variable pair, the app computes a global association measure, a significance test, and local diagnostics adapted to the variable types; users can then filter an interactive association network and inspect retained edges through harmonized pair plots.

Choose a dataset

Select variables
and controls

Compute pairwise
associations
and tests

Filter the
network view

Open local
pair plots

All example figures on this poster are based on ESS 2011 (European Social Survey), Belgium.

2 - Association Measures

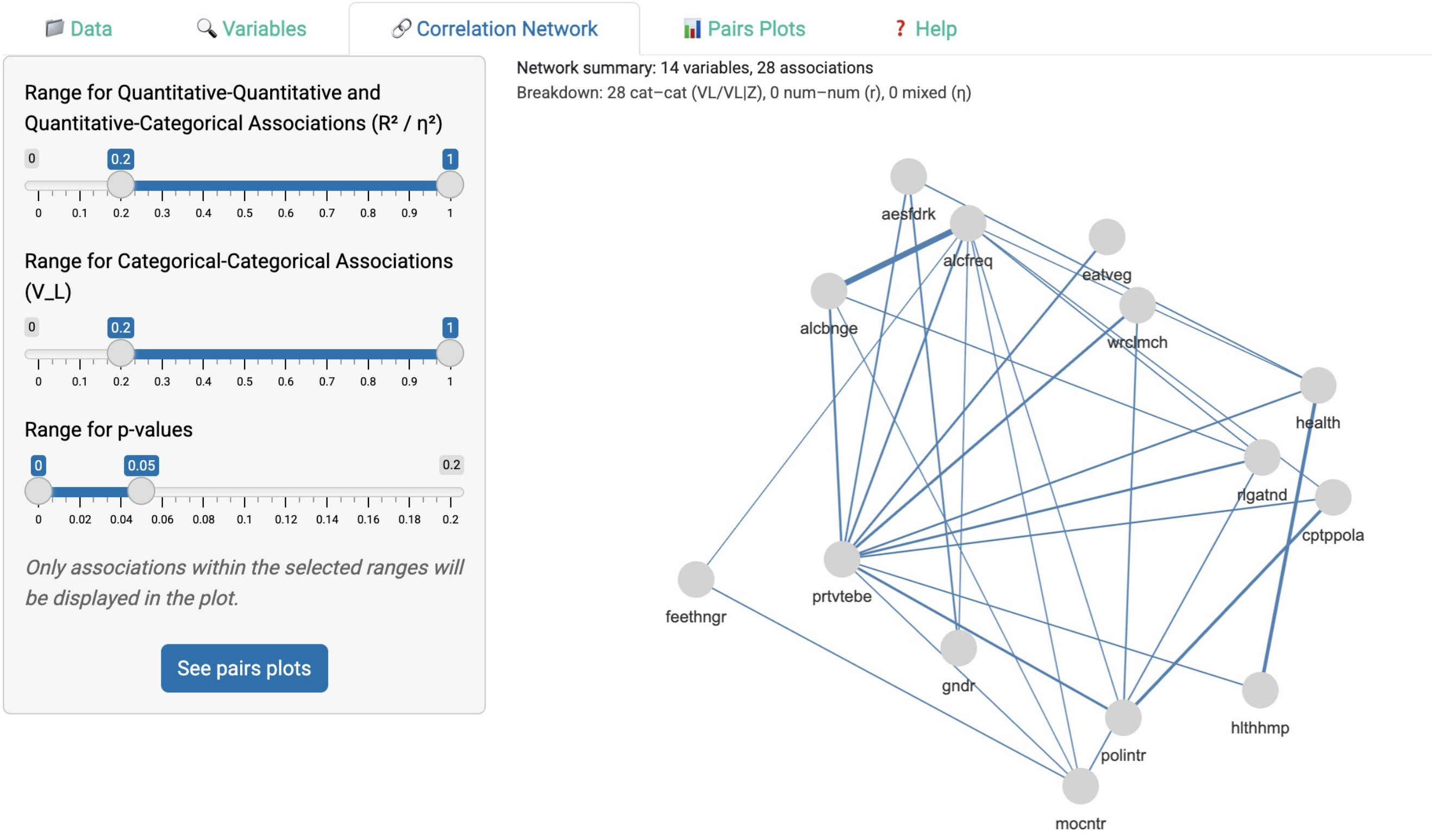
Variable types	Unconditional measure	Conditional measure
Numerical–numerical	R^2 (Pearson)	$R^2_{X \cdot Z}$ (partial)
Numerical–categorical	η^2 (ANOVA)	$\eta^2_{X Z}$ (ANCOVA)
Categorical–categorical	V_L	$V_{L Z}$

Categorical–categorical specification

$$V_L = \sqrt{1 - \exp\left(-\frac{G^2}{n}\right)}$$
$$G^2 = 2(\ell_1 - \ell_0)$$

- ℓ_0 is the fitted log-likelihood under the null model of (conditional) independence; ℓ_1 is the fitted log-likelihood under the alternative model allowing association between the two categorical variables.
- Without controls, G^2/n equals twice the empirical mutual information; V_L is therefore a bounded information-based measure of association, corresponding to Linfoot's coefficient at the sample level.

3 - Filtered Network

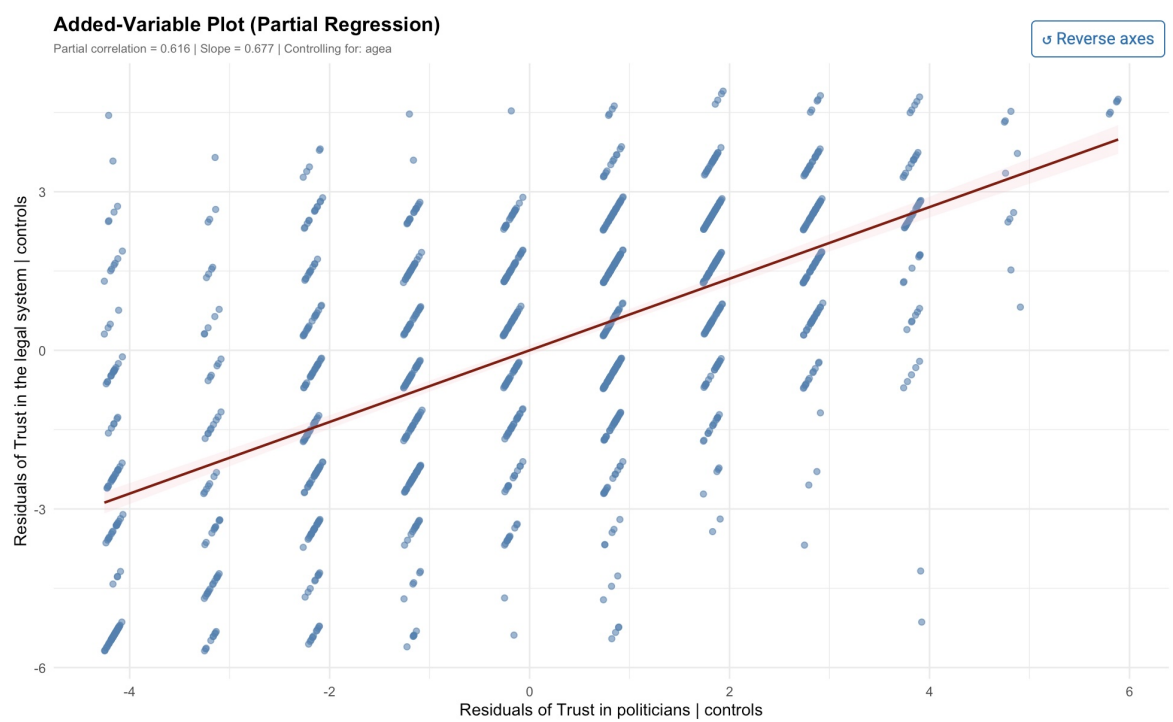


- Nodes** are selected variables; **edges** are retained pairwise associations
- Range sliders restrict the displayed graph by **p-value**, by R^2/η^2 for numerical–numerical and numerical–categorical pairs, and by V_L for categorical–categorical pairs

4 - Pair Plots

4.1 Numerical–numerical

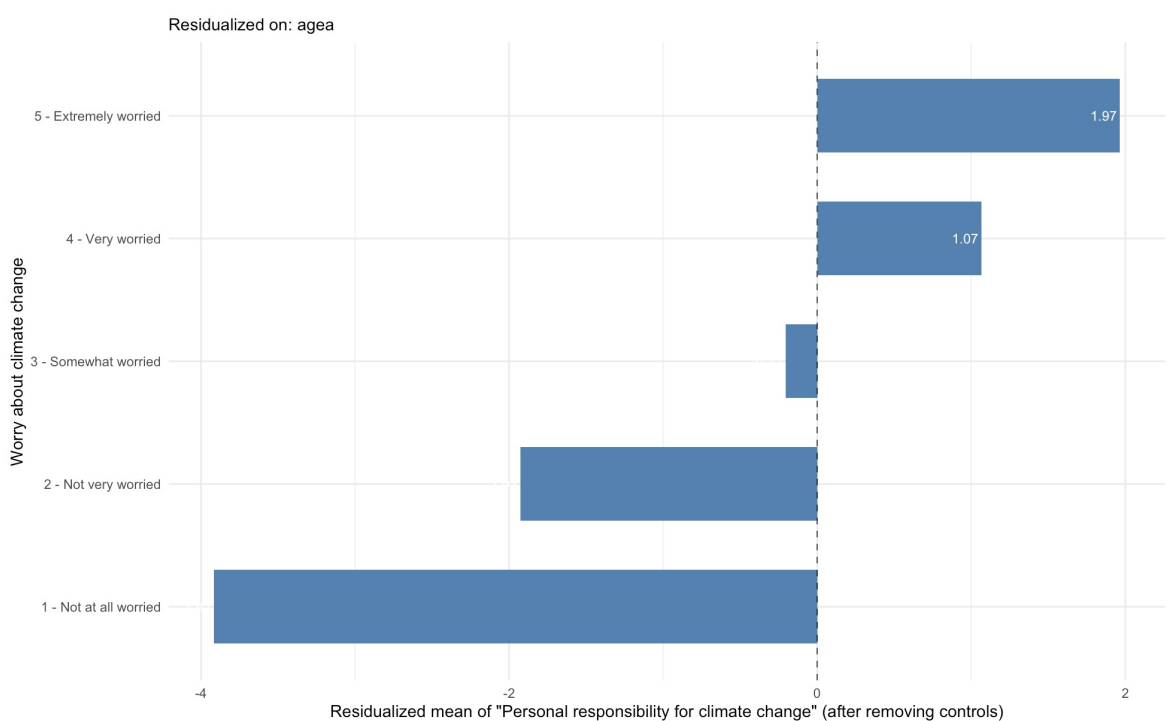
- Scatter plot without controls; added-variable plot (scatter plot of residuals of Y and X on Z) with controls



Without controls, the slope reflects the sign of r ; with controls, the added-variable slope reflects the sign of the partial correlation

4.2 Numerical–categorical

- Raw group means without controls; residualized group means with controls



With controls, bars show means of the residualized outcome after regressing Y on Z

4.3 Categorical–categorical

- The pair plot is a local association table built from observed counts O_{ij} and expected counts $E_{ij}^{(0)}$ under the null model
- Cell values show $D_{ij} = O_{ij} - E_{ij}^{(0)}$: D_{ij} measures the departure from the null model in count units
- Cell colors are based on $R_{ij} = (O_{ij} - E_{ij}^{(0)})/\sqrt{E_{ij}^{(0)}}$: R_{ij} standardizes this departure relative to its expected variability. A stronger red color indicates that the observed count is substantially higher than expected under independence. A stronger blue color indicates that the observed count is substantially lower than expected under independence.

Unconditional categorical association

VL = 0.405 | p-value = 3.83e-14
Cell values show D = O - E0. Colors are based on Pearson residuals R: red = over-represented, blue = under-represented, darker = stronger.
Large table detected (5x16 = 80 cells). Showing the best 5x7 submatrix selected from likelihood-ratio cell contributions (76.2% of total G² contribution).

Worry about climate change	CD&V	Ecolo	Groen!	MR	N-VA	PS	Vlaams Belang
1 - Not at all worried	-1.76	-1.02	-1.65	-3.03	0.12	-1.68	4.29
2 - Not very worried	-4.26	-7.21	-8.77	8.69	2.05	8.10	3.97
3 - Somewhat worried	2.89	-8.95	-17.37	-5.43	16.20	-0.79	16.79
4 - Very worried	5.15	13.18	16.99	4.27	-14.51	-5.32	13.63
5 - Extremely worried	-2.03	3.99	10.80	-4.51	-3.86	-0.30	-6.41

Local associations between the variables 'Worry about climate change' and 'Party voted for during the last elections'